Recitation 11

# Exercise 1

Consider the scalar linear discrete-time system

$$x(k+1) = (2+a)x(k) + u(k) \tag{1}$$

with unknown parameter $a \in \mathbb{R}$, $|a| \leq a_u$. The system is constrained as follows:

$$x(k) \in \mathcal{X}, \qquad u(k) \in \mathcal{U}, \qquad \mathcal{X} := [-5, 5],\ \mathcal{U} := \mathbb{R}.$$

1. Provide the disturbance set $\mathcal{W}$ such that for $w(k) \in \mathcal{W}$

$$x(k+1) = 2x(k) + u(k) + w(k) \tag{2}$$

captures the uncertainty due to not knowing $a$.

In the following, consider system (2) with $|w(k)| \leq 2$.

2. Calculate a linear state feedback gain $K_{\mathcal{E}}$, such that the minimal robust invariant set for system (2) under $u(k) = K_{\mathcal{E}} x(k)$ is given by $\mathcal{E} = [-3, 3]$.
3. Assume $K_{\mathcal{E}} = -2$. Compute the minimal robust invariant set $\mathcal{E}$ for system (2) under the control law $u(k) = K_{\mathcal{E}} x(k)$ as well as $\bar{\mathcal{X}} = \mathcal{X} \ominus \mathcal{E}$ and $\bar{\mathcal{U}} = \mathcal{U} \ominus K_{\mathcal{E}} \mathcal{E}$.

*Hint 1:* $[a, b] \oplus [c, d] = [a+c, b+d]$, $[a, b] \ominus [c, d] = [a-c, b-d]$

*Hint 2:* $\sum_{i=0}^{\infty} q^i = \begin{cases} \frac{1}{1-q}, & |q| < 1, \\ \infty, & \text{otherwise.} \end{cases}$

# Exercise 2

Consider the linear system

$$x(k+1) = \underbrace{\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}}_{A} x(k) + \underbrace{\begin{pmatrix} 0.5 \\ 1 \end{pmatrix}}_{B} u(k) + w(k)$$

with constraints $x(k) \in \mathcal{X} := \{x \in \mathbb{R}^2 : [0\ 1]x \leq 2\}$, $u(k) \in \mathcal{U} := \{u \in \mathbb{R} : |u| \leq 1\}$, and $w(k) \in \mathcal{W} := \{w \in \mathbb{R}^2 : ||w||_\infty \leq 0.1\}$. Let $\ell(z, v) = (1/2)z^\top Q z + (1/2)v^\top R v$ with $Q = I$, $R = 0.01$. Design and implement a tube-based MPC controller for the system as outlined below.

i) Determine a valid $K_{\mathcal{E}}$ and the corresponding minimal robust positively invariant set $\mathcal{E}$.

ii) Determine a valid terminal controller $K_f$ and a corresponding terminal set $\mathcal{X}_f$.

iii) Implement an MPC algorithm for the problem above with prediction horizon $N = 5$, in particular think of the following aspects:

- Which are the variables to optimize over?
- How can the constraints be implemented?
- What can you observe about the initial state $z_0$ at each iteration?

iv) Approximately compute the feasible set of the MPC and plot it for different planning horizons

## Exercise 3

Consider the nominal linear system

$$x(k+1) = Ax(k) + Bu(k), \tag{3}$$

with $(A, B)$ controllable, and the uncertain linear system

$$x(k+1) = Ax(k) + Bu(k) + w(k), \tag{4}$$

with the same $A, B$ matrices and unknown but bounded disturbances $w \in \mathcal{W} \subset \mathbb{R}^n$ with $0 \in \mathcal{W}$. Both systems (3) and (4) are subject to state constraints $x(k) \in \mathcal{X}$ and input constraints $u(k) \in \mathcal{U}$, for all $k \geq 0$.

Consider the standard nominal MPC controller $\kappa$ defined by the optimization problem $P_{S.1}$ as introduced in the supplementary material S.1, satisfying Assumptions S.1-1, S.1-2, and S.1-3, for system (3), and the standard tube-based MPC controller $\kappa_{\text{tube}}$ defined by the optimization problem $P_{S.2}$ as introduced in the supplementary material S.2, satisfying Assumptions S.2-1, S.2-2, and S.2-3, for system (4).

| | True | False |
|---|---|---|
| The robust tube-based MPC problem $P_{S.2}$ is recursively feasible for all disturbances $w(k) \in \mathcal{W}$. | ☐ | ☐ |
| Suppose $P_{S.2}$ is feasible for $x(k)$ with optimal input sequence $V^*(x(k))$. Is $V^*(x(k))$ guaranteed to be a feasible input sequence for $P_{S.1}$? | ☐ | ☐ |
| Consider the closed-loop system (4) under $u(k) = \kappa_{\text{tube}}(x(k))$. It holds that $\sum_{i=0}^{\infty} \ell(x(i), u(i)) < \infty$. | ☐ | ☐ |
| If $\mathcal{U}$ is a polytope and $\mathcal{E}$ is a polytope, then $U \ominus K_{\mathcal{E}}\mathcal{E}$ is a polytope if it is a non-empty set. | ☐ | ☐ |
| Consider a minimal robust invariant set $\mathcal{E} \subseteq \mathcal{X}$ for system (4) under the controller $u(k) = K_{\mathcal{E}}x(k)$ with $K_{\mathcal{E}}\mathcal{E} \subseteq \mathcal{U}$, and replace the terminal constraint in $P_{S.1}$ with $x_N \in \mathcal{E}$. The MPC problem $P_{S.1}$ is recursively feasible for the closed-loop system (3) with $u(k) = \kappa(x(k))$. | ☐ | ☐ |
| Consider a minimal robust invariant set $\mathcal{E} \subseteq \mathcal{X}$ for system (4) under the controller $u(k) = K_{\mathcal{E}}x(k)$ with $K_{\mathcal{E}}\mathcal{E} \subseteq \mathcal{U}$, and replace the terminal constraint in $P_{S.1}$ with $x_N \in \mathcal{E}$. The origin is guaranteed to be an asymptotically stable equilibrium for the closed-loop system (3) with $u(k) = \kappa(x(k))$. | ☐ | ☐ |